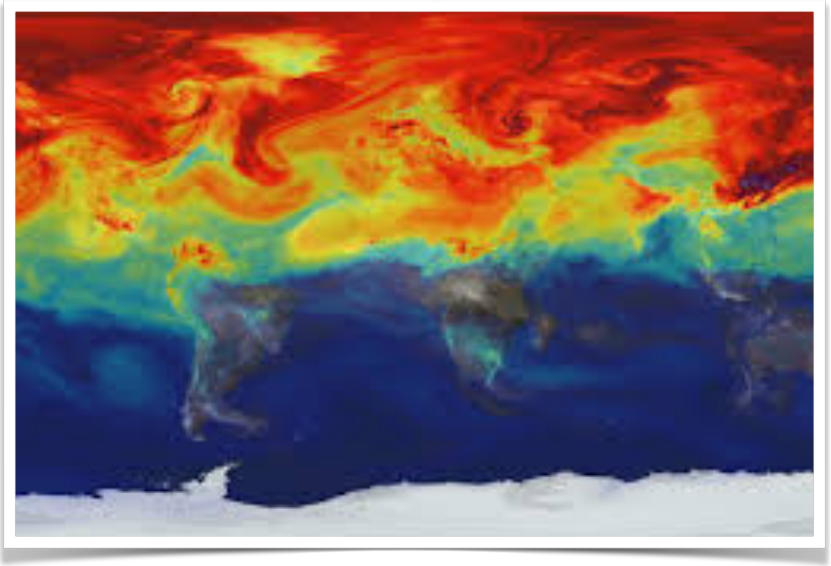
MAD AAG | American Association of Geographers



Geography Matters:
MAD-AAG 2024

The Department of Geography & Environmental Systems at the University of Maryland Baltimore County (UMBC) Welcomes Everyone to the 2024 Middle Atlantic Division (MAD), American Association of Geographers Annual Meeting.

8:00 - 9:00
Arrivals, Registration, Breakfast, Welcome

9:15 - 10:15 Keynote Talk

"Punitive Psychiatry: The Prosaic Geographies of the U.S. Security State"

Dr. Nicole Nguyen, Scholar Activist, Feminist Geographer, & Professor of Criminology, Law, & Justice, University of Illinois-Chicago

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8:00 - 9:00 Arrivals, Registration, Breakfast, Welcome

9:15 - 10:15 Keynote Talk

10:30 - 12:00 Concurrent Sessions A & B

12:15 - 1:15 LUNCH and MAD Business Meeting

1:30 - 3:15 Concurrent Sessions C & D

3:30 - 6:00 Annual MAD GeoBowl



The intersection of geography and learning is more urgent than ever as we all experience change and disruption not just in "real" time but in "Space-Time" at increasingly dizzying speeds

Session A: 10:30 - 12:00

Sustaining Growth, Pursuing Justice, and Community Engagement

Blue Bonds as a Catalyst for Blue Economic Growth in the Northeastern USA

Keywords: Blue bonds; Sustainable Finance; Innovative Financing; Blue Economy

Author(s): Hilda Afeku-Amenyo, Montclair State University

Pankaj Lal, *Montclair State University*

Growth of solar energy projects in Northumberland County, Pennsylvania

Keywords: Solar panels, Growth, Northumberland County, PA, Timeline, governmental policies, rural community, economy.

Author(s): Hasiba Hasin Royesh, *Bucknell*

Iceland's transition to a clean energy economy

Keywords: Iceland, clean energy infrastructure, hydro-electric, geothermal, scale, policy

Author(s): Gina Bloodworth, Salisbury University

Weed Warriors and Climate Champions: Green Labor at the Confluence of Nature and Identity

Keywords: situated urban political ecology, community engagement, nature, conservation, sustainability, environmental justice, green space, non-governmental organizations

Author(s): William Kaselow, *University of Maryland, Baltimore County*

Session B: 10:30 - 12:00 Resisting Gendered/Racial Environmental Injustice; Embodied Migration; and Sustainable Urbanism

The gendered burden of environmental responsibility for lead poisoning in Baltimore, MD

Keywords: environmental justice, motherhood, feminist geographies, lead poisoning

Author(s): Samantha DiNatale, University of Maryland, Baltimore County

Challenging Disasters Through Black Geographies: Deciphering the Plot of Resilience in West Baltimore, Maryland, USA

Keywords: Disasters, Environmental Hazards, Racial Violence, Black Geographies, Disaster Preparedness, Resistance

Author(s): Nyla Howell, Rutgers University - New Brunswick
Dawn Biehler, PhD, University of Maryland Baltimore County
Baltimore's Coalition for Positive Change
Assata Hanif, *University of Maryland Baltimore County*

Venezuelan Women's Migratory Movements to the USA: Perceptions of Agency, Land, Community, and Government. Preliminary Results

Keywords: Identity, citizenship, community building, racial capitalism, violence, and border and body.

Author(s): Laura Torres-Vargas, UMBC, GES

La ville nouvelle de Diamniadio : les défis de la gouvernance d'un espace de production, de distribution et de consommation. / The New Town of Diamniadio: The Challenges of Governing a Space of Production, Distribution and Consumption.

Keywords: Ville nouvelle, gouvernance urbaine, urbanisation, relations villes-campagnes, Diamniadio/New town, urban governance, urbanization, town-countryside relations, Diamniadio

Author(s): Modou Ndiaye, *Institut Fondamental d'Afrique Noire Cheikh Anta*

Session C: 1:30 - 3:15 Cripping Academic Time: Navigating, Researching, and Refusing Neoliberal Compression

Humanizing Higher Ed: Epistemic Resistance, Vicarious Resilience, and Neuroqueer Academic Space

Keywords: neuroqueer, critical geographies, community care, epistemic resistance, storytelling, vicarious resilience, crip time, neurodivergence, digital geographies, leverage points

Author(s): Tal Garbo, *University of Maryland, Baltimore County*

Neurodivergent Education: Navigating University Demands as a Neurodivergent Faculty Member

Keywords: Neurodivergence, University Faculty, Support Services

Author(s): Jenni Connell, Towson University

Zine and the Art of Mind Melting

Keywords: Care ethics, zines

Author(s): Leah Franklin, *Towson University*

A postscript on rhythmic praxis

Keywords: slow scholarship; crip time; academic labour

Authors: *Kelsey Hanrahan, Towson University*

Slow(ed) Scholarship: On Doing Slow Vs. Being Slowed

Keywords: slow scholarship, neurodivergence, crip time, disability justice, feminist geography, academia

Author(s): E.C. Kaufman, *Towson University*

Carrie Chennault, Colorado State University
Hanieh Molana, California State University, Sacramento

Geography Matters
Where Learning Comes Naturally

Geography provides an education that everyone deserves. There is no silver bullet to resolve global challenges but geography provides the intellectual glue that binds together multidisciplinary insights to help us create the worlds we envision.

Session D: 1:30 - 3:15 Smoothing Spatio-Temporal, Urban-Rural, Geological, and Green Disparities and Disturbances

Marie Tharp and her Contributions to Geography

Keywords: Mid-Atlantic Ridge, Marie Tharp, Continental Drift, Seafloor Spreading

Author(s): Rebecca S Lee, Frostburg State University

Greenspaces for Whom? Urban Tree Canopy as a Lens into Equity in Baltimore

Keywords: climate, justice, urban greening, history, trees, tree plantings

Author(s): Brandon Miller

Monitoring Emergent Tropical Tree Density to Landscape Changes Using High-Resolution Satellite Imagery and Machine Learning: A Case Study with Dipteryx panamensis

Keywords: Tropical Forests, Emergent trees, Remote Sensing

Author(s): Felipe Saad, *UMBC-GES*

Matthew Fagan, *UMBC - GES*

Rohit Mukherjee, *PNNL*

Exploring Geographic and Temporal Disparities in Healthcare Services

Keywords: Spatial disparity, Spatial-temporal discrepancy, Clustering analysis, Geographic scales, South Korea

Author(s): Hyun Kim, *University of Tennessee, Knoxville, USA*

Changwha Oh, *University of Tennessee, Knoxville, USA*

Wanhee Kim, *University of Tennessee, Knoxville, USA*

Toward an Urban-Rural Continuum: Settlement Classification Research at the Census Bureau

Keywords: Urban, Rural, Settlement Types, Census Bureau

Author(s): Michael Ratcliffe, *US Census Bureau*

ABSTRACTS

Blue Bonds as a Catalyst for Blue Economic Growth in the Northeastern USA

Keywords: Blue bonds; Sustainable Finance; Innovative Financing; Blue Economy

Author(s): Hilda Afeku-Amenyo, Montclair State University

Pankaj Lal, *Montclair State University*

This research investigates the role of blue bonds as a transformative financial instrument in fostering Blue Economy growth in the Northeastern United States. As designed, blue bonds will fund ocean-related projects that can align financial returns with environmental sustainability. The study shall examine how these bonds can finance diverse marine and coastal activities, such as sustainable fisheries, marine conservation, and coastal resilience initiatives. This study will pursue the economic viability of blue bonds through various case studies in different parts of the world, analyzing against the relevant policy frameworks to determine how well they address regional environmental objectives.



The research objectives will study the prospects for blue bonds regarding their potential for sustainable economic growth, identify problems, including financial, regulatory, and environmental constraints, to the successful issue of blue bonds, and assess the compatibility of blue bonds under current policy frameworks.

Early evidence suggests that blue bonds can widen sources of financing for Blue Economy activities, contribute to job creation, and induce innovation in ocean technologies. However, a multiplicity of regulatory regimes, nascent market conditions, and perceived investment risks will all need to be tackled in order to increase investor confidence. Conclusively, this research allows the determination of how blue bonds can, in fact, support the financing of ocean-based sustainable development in the Northeastern USA and provide valuable input to policymakers, investors, and stakeholders. It advocates for development within a Blue Economy using innovative mechanisms for financing fiscal sustainability while protecting marine ecosystems.

Growth of solar energy projects in Northumberland County, Pennsylvania

Keywords: Solar panels, Growth, Northumberland County, PA, Timeline, governmental policies, rural community, economy.

Author(s): Hasiba Hasin Royesh, *Bucknell*

In this research, I explore the growth of solar energy projects in Northumberland County, Pennsylvania, mapping out key developments, setbacks, and local responses over the past several years. Focusing on the transition toward renewable energy in rural areas, I analyze major projects, like the U.S. Department of Agriculture's recent $5.2 million investment aimed at supporting solar initiatives for farmers, small businesses, and local communities across Pennsylvania.

This study brings to light the involvement of critical stakeholders, including government agencies and private companies, and examines how county policies are evolving to accommodate new solar installations. The research also delves into the challenges faced by Northumberland County—ranging from zoning and environmental regulations to public resistance—as local leaders attempt to balance the benefits of economic investment with the preservation of community values and landscapes. By piecing together information from policy documents, media coverage, and community input, this project provides a detailed view of Northumberland County's efforts to adopt clean energy, while reflecting the broader realities and complexities of implementing solar energy in rural America.



Iceland's transition to a clean energy economy

Keywords: Iceland, clean energy infrastructure, hydro-electric, geothermal, scale, policy

Author(s): Gina Bloodworth, *Salisbury University*

Iceland stands on the global forefront of efforts to transition national economies towards a carbon-neutral footprint, and now can boast 100% renewable electricity generation. Currently, one quarter of Iceland's electricity comes from hydro-electric sources while three fourths comes from geothermal sources. Similarly, Iceland's capital city has integrated geothermal heating throughout the city. How Iceland managed this transition shows both potential paths forward for others, and the importance of scale as a logistical challenge. To consider how other nations might learn from Iceland's recent accomplishments, a comparative case study is being developed between Iceland and the U.S. In America, just 11% of our energy comes from renewable sources, and if that percentage must increase the obvious questions are how? When? At what cost? Can the U.S. use Iceland's pathway as a roadmap for our own transition to a clean energy economy? This is a comparative case study of clean energy infrastructure and policy for the U.S and Iceland. A comparative analysis of renewable energy sources, pathways, KW generation, distribution and consumer strategies, I hope to show where the road ahead looks promising, or problematic in terms of physical infrastructure. The policy component looks first at Iceland's recent policy on renewable energy compared with the U.S. But given our own nation's political shift, we must also consider what infrastructure is predictable [established now] in the U.S. or unpredictable [not yet fully established] in the coming decade within our nation

The gendered burden of environmental responsibility for lead poisoning in Baltimore, MD

Keywords: environmental justice, motherhood, feminist geographies, lead poisoning
Author(s): Samantha DiNatale, University of Maryland, Baltimore County

In accordance with a push for increased intersectional approaches in critical environmental justice scholarship, there is growing awareness of the important role of gender in shaping experiences of injustice. This presentation takes a gendered lens in the exploration of who has been historically framed as responsible for the children's lead poisoning epidemic in Baltimore, Maryland. I use archival methods to investigate how maternal responsibility was constructed in newspaper publications, often possessing underlying intersections with race and class. Findings reveal that gendered assumptions often place the burden of lead exposure on mothers, overlooking broader contributors and inaction of state entities. This analysis highlights the need to challenge these narratives and reframe environmental justice to shift away from the individualization of responsibility, contributing to more equitable environmental health discourse.



Challenging Disasters Through Black Geographies: Deciphering the Plot of Resilience in West Baltimore, Maryland, USA

Keywords: Disasters, Environmental Hazards, Racial Violence, Black Geographies, Disaster Preparedness, Resistance
Author(s): Nyla Howell, Rutgers University - New Brunswick
Dawn Biehler, PhD, University of Maryland Baltimore County
Baltimore's Coalition for Positive Change
Assata Hanif, *University of Maryland Baltimore County*

Although climate-induced hazards and disasters disproportionately impact Black communities, scholarly discourse on these issues has historically been dominated by those outside the affected communities. To address the role of systemic racism in unequal disaster experiences, scholars must adopt approaches that center Black experiences and knowledge. This research delves into the intersections of Black geographic thought and disaster studies to support Harlem Park, a predominantly and historically Black community in Baltimore, Maryland, in their goal of creating a community climate resilience program aligned with abolitionist climate justice.

Weed Warriors and Climate Champions: Green Labor at the Confluence of Nature and Identity

Keywords: situated urban political ecology, community engagement, nature, conservation, sustainability, environmental justice, green space, non-governmental organizations

Author(s): William Kaselow, *University of Maryland, Baltimore County*

Political ecological perspectives establish that environmental non-governmental organizations (eNGOs) are influential actors in the production of nature; emphasizing the role of multinational eNGOs in reproducing colonial power dynamics through the enclosure of land and reframing the erasure of traditional land relations as conservation and green development. Urban political ecology translates these dynamics to the city through green frontiers, and situated approaches complicate our understanding of eNGOs by recognizing that they provide a range of localized benefits to communities including education, recreation, and resource distribution. This research employs community-engaged methods to examine the influence of Baltimore's environmental non-governmental organizations (eNGOs) on public conceptions of nature, conservation, sustainability, and environmental justice (EJ). Leaning on insights from Black Feminist and Indigenous geographies, I pair concepts of metabolism and identity formation to understand how eNGOs' public programs reflect the interrelated influences of power and place. Preliminary fieldwork suggests that eNGOs can simultaneously uphold and resist inequitable power relations, and subsequently, that eNGOs are implicated in the tension between conservation and EJ in Baltimore. In this paper, I put the mutual construction of identity and place in conversation with metabolic rift to examine how power is reproduced and resisted in urban approaches to conservation and sustainability. I argue that the labor deployed by eNGOs through public programs represents distinct metabolic relationships and that the discourses which characterize such programs encourage the formation of identities relative to particular, value-laden, conceptions of nature; e.g. Weed Warriors, stewards, champions, etc.

La ville nouvelle de Diamniadio : les défis de la gouvernance d'un espace de production, de distribution et de consommation. / The New Town of Diamniadio: The Challenges of Governing a Space of Production, Distribution and Consumption.

Keywords: Ville nouvelle, gouvernance urbaine, urbanisation, relations villes-campagnes, Diamniadio/New town, urban governance, urbanization, town-countryside relations, Diamniadio

Author(s): Modou Ndiaye, *Institut Fondamental d'Afrique Noire Cheikh Anta*

Located 30 km east of the capital, Diamniadio welcomes an increasing number of residents and voracious modes of consumption of urban services. The fundamental question that guided our research is whether the rapid and blind growth that prevails in Diamniadio is the result of real estate speculation that excludes the most deprived. It is imperative to resolve the problems posed by the urbanization of the new city in a collaborative manner by relying on the demands expressed by the populations of the neighboring areas. This approach aims to expose the limits of the centrality of the key issues of the governance of Senegalese cities both in terms of the means mobilized and the founding philosophies of the options. In particular, it is a question of replacing the Diamniadio new city project in the ongoing debate on the sustainability of urban centers and its relevance. However, the accelerated pace of urbanization in Diamniadio requires the establishment of a new governance framework to adapt to the new faces of the city, improve its management, and strengthen collaboration between urban and rural areas.

Venezuelan Women's Migratory Movements to the USA: Perceptions of Agency, Land, Community, and Government. Preliminary Results

Keywords: Identity, citizenship, community building, racial capitalism, violence, and border and body.

Author(s): Laura Torres-Vargas, UMBC, GES

Currently, migration in the world is a reality that affects many, in some cases it affects us personally and in other people close to us. Is necessary to recognize that migratory movements are not always voluntary and on occasions are a response to social, economic, political problems, among others. The case of Venezuelan women, these problems have influenced the decision to migrate; it has led them to begin journeys of migration to countries such as Colombia, Panama, and, on many occasions the USA as their final destination. In these migratory processes, Venezuelan women experience different realities that change their perspectives of what surrounds them and their agency in non-binary ways; as well as allowing them to internalize experiences that can challenge their idea of community, government, identity, and border and body. This poster presents the methods used to gather data from Colombia, Panama, and the USA answering to how Venezuelan women have experienced their migratory movement and how external factors such as job opportunities, education, sex, economic solvency, and age influence their perception of the social, border and body, and governmental relationships.



Humanizing Higher Ed: Epistemic Resistance, Vicarious Resilience, and Neuroqueer Academic Space

Keywords: neuroqueer, critical geographies, community care, epistemic resistance, storytelling, vicarious resilience, crip time, neurodivergence, digital geographies, leverage points

Author(s): Tal Garbo, *University of Maryland, Baltimore County*

Although epistemic resistance, liberatory community care, and academic interests are common and overlapping facets of neuroqueer/disabled populations, not only is this praxis made nearly unsurvivable by capitalistic reflections of labor expectations in academia, but theoretical and physical space is rarely made with or for us. Recent years show an increase in disability visibility as well as scholarship by disabled academics; and yet, the pathology paradigm and capitalist devaluations of crip time and other disabled modes of embodied resistance persist in shaping spaces and practices that surround us. What can be done? Mapping disabled stories, specifically highlighting neuroqueer experiences in academic spaces and where they diverge, can provide leverage points in furthering the current paradigmatic shift being undertaken by many disabled scholars and feminist geographers. As many neurodivergent people are natural story weavers and anecdotal communicators, taking a triangulated methodological approach in the form of autoethnography surrounding ongoing participation in both formal and informal academic spaces, as well as ethnographic research involving observational study and survey work (with emphasis on varied storytelling options for participants) to

encompass the variety of communication needs and preferences among diverse bodyminds and how they express their spatial experiences is a necessity. As we gather data on how neuroqueer/disabled scholars move through digital and physical geographies of academia, and move away from formal academia despite ongoing academic pursuits/interests, we can identify further leverage points, grow vicarious resilience, and develop stronger collective understanding of how to navigate these spaces while still centering our bodyminds.



Neurodivergent Education: Navigating University Demands as a Neurodivergent Faculty Member

Keywords: Neurodivergence, University Faculty, Support Services

Author(s): Jenni Connell, Towson University

As our society becomes more aware of the importance of mental health and gains a greater understanding of different neurodivergent identities such as autism or ADHD, how we provide support becomes increasingly relevant. Often, the focus is on students. This project aims to examine educators instead. My research explores the supports available to neurodivergent university faculty and the impacts available supports have on their career contentment and mental health. Data gathering is underway using one on one semi-structured interviews, the PVM of drawing, or a combination of the two. I lean on some concepts from the social model of disability, acknowledging that there are societal practices that are the root cause of disabling circumstances, while not demonizing all medical and social care that may be needed or desired. I seek to bring awareness to the needs of neurodivergent individuals in academia, including those who are not receiving any supports due to a lack of disclosure of their status for personal or professional reasons. Based on initial findings, I argue that there is a need for additional supports for neurodivergent faculty in order to meet the allistic demands of university employment.

Zine and the Art of Mind Melting

Keywords: Care ethics, zines

Author(s): Leah Franklin, *Towson University*

This semester I entered into an independent study on feminist ethics of care, wanting to learn more about care but also knowing I needed to care for myself a bit. I was hesitant to engage in work that would be intense and disruptive. Towards these ends, we decided on a creative method for documenting my thoughts on the readings -- a zine -- and we put together a preliminary set of readings that would be open to change, and responsive to my own interests and reactions.

Then my mind melted.

This presentation is a reflection of my experiences in engaging with a body of literature that was disruptive physically and emotionally. Working on my zine provided the space to explore how I was upending my understanding of what it means to care and what it means to care for my brain, to allow it to work the way it wants to work. This slower process is opening space to sit with ideas and interrogate my relationship to these ideas (in a way that is traditionally perceived as inefficient and unproductive) and it has opened up a new possibility of what learning could be.



A postscript on rhythmic praxis

Keywords: slow scholarship; crip time; academic labour

Authors: *Kelsey Hanrahan, Towson University*

This presentation is a postscript to a paper I presented in August 2024 with my collaborator Emily Billo. Over the last five years, we have worked together to build a rhythmic praxis that allow us to support and care for each other while resisting the neoliberal rhythms that dominate academic work in our institutions and disciplines, rhythms that feel exhausting and depleting. The fast-paced rhythms reproduced in writing cultures emphasize continual productivity, individuality, and transactional interactions while a rhythmic praxis requires everyday fine tuning in relation to shifting emotional and visceral responses to our circumstances, our silences, the pace and tempos of the processing of ideas. We concluded that paper with questions around what it would mean to translate our rhythmic praxis—shaped and practiced in collaborative space and co-authorship—into our separate research projects which had largely been put on hold during the first few years of the COVID pandemic. In this presentation I reflect on my recent embodied experiences as I reimagine this praxis for my independent research and writing, turning to work on crip time, chronic illness, and neurodiversity as I work to expand how I can imagine my bodymind, academic labour, and the rhythms and temporalities that we create.

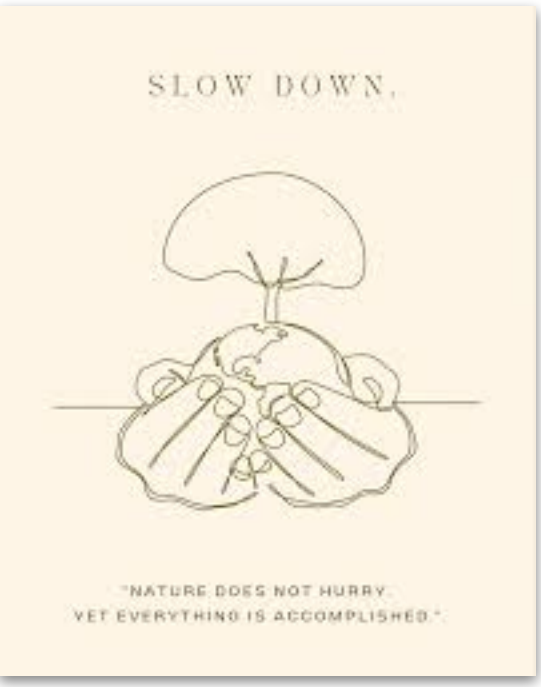
Slow(ed) Scholarship: On Doing Slow Vs. Being Slowed

Keywords: slow scholarship, neurodivergence, crip time, disability justice, feminist geography, academia

Author(s): E.C. Kaufman, *Towson University*

Carrie Chennault, Colorado State University
Hanieh Molana, California State University, Sacramento

This paper is based on an experiment in knowledge co-production: a recorded conversation between three early career feminist critical geographers, facing disabling conditions and the pressure of neoliberal time in academia. We connect literature on slow scholarship, time, disabilities, and neurodivergence with our personal journeys navigating health, neurodivergence, and time in North American academic institutions. In rethinking what "slowing down" and refusal mean from the perspective of an 'already slowed body,' this paper challenges silencing taboos and encourages more conversations among scholars at different career stages who face varying levels of precarity and 'slowed' by different barriers and conditions. However, the conversations and stigmatized sharing we encourage is built on the foundation that such talks, individualizing solutions, universal design, or simply 'more awareness' are never the end goal. Rather, that goal is changing the material conditions of inequity, stress, and physical pain experienced by scholars at the intersections we describe or inhabit. A commitment to slow scholarship in feminist and critical geographies demands a commitment to those who wrestle with time and health in academia, and to those who inhabit the paradox between slowing down and keeping up.



Exploring Geographic and Temporal Disparities in Healthcare Services

Keywords: Spatial disparity, Spatial-temporal discrepancy, Clustering analysis, Geographic scales, South Korea

Author(s): Hyun Kim, *University of Tennessee, Knoxville, USA*
Changwha Oh, *University of Tennessee, Knoxville, USA*
Wanhee Kim, *University of Tennessee, Knoxville, USA*

Spatial disparities in healthcare services have become a core research focus across fields such as public health, geography, and GIS. Despite various methodological advancements, most approaches remain empirical, shaped by research objectives, data types under a certain geographic scale. This research introduces novel research methods for analyzing spatial disparities in healthcare services via spatial discrepancy indexes and a time-series clustering approach for various geographic and temporal dimensions. As case studies, we examine the provision of (1) Essential Primary Care Services during pandemic periods in South Korea, and (2) Pediatric Clinics, which have experienced critical declines between 2000 and 2020 in Seoul Metropolitan Area. Our proposed disparity indexes employ straightforward yet effective metrics to measure mismatches between service provision and demand, and service stress level in serving healthcare demands at different geographic scales. In addition, a dynamic time warping clustering method is employed to capture the changes in the supply of pediatric clinics in a longitudinal perspective. Our framework provides a robust approach for analyzing healthcare access with multiple spatial and temporal dimensions considered, offering a unique perspective on healthcare disparity issues.

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Marie Tharp and her Contributions to Geography

Keywords: Mid-Atlantic Ridge, Marie Tharp, Continental Drift, Seafloor Spreading

Author(s): Rebecca S Lee, Frostburg State University

Through Marie Tharp's diligent research in mapping the ocean floor, we came to better understand the world that surrounds us and the history behind it. Alfred Wegener's continental drift hypothesis was ridiculed for the many years it stayed shunned by the scientific community. It wasn't until Marie Tharp and Bruce Heezen worked together to propose and put together a mechanism that supported the continental drift hypothesis was it then accepted. Where Alfred Wegener has always supported the idea of convection currents powering the movement of tectonic plates, Marie Tharp discovered a 40,000-mile-long ridge that supported the concept of seafloor spreading. Without this concept, the fundamentals of geology may not have been laid.

Toward an Urban-Rural Continuum: Settlement Classification Research at the Census Bureau

Keywords: Urban, Rural, Settlement Types, Census Bureau

Author(s): Michael Ratcliffe, *US Census Bureau*

In this presentation, I discuss research at the Census Bureau focused on developing a continuum of settlement categories for use in analysis and data presentation. Practitioners generally agree that a dichotomous approach to defining urban and rural no longer adequately supports the needs of analysts, researchers, and policy makers. Instead, classification of communities along an urban-rural continuum facilitates analysis of characteristics and conditions in communities that differ in terms of settlement pattern, settlement intensity, and proximity to/distance from other population centers. After a brief review of the Census Bureau's and other agencies' urban and rural classifications, I will present options currently under consideration, including potential alignment with the Global Human Settlement Layer/Degree of Urbanization, which was adopted by the United Nations to facilitate cross-country comparisons.

Greenspaces for Whom? Urban Tree Canopy as a Lens into Equity in Baltimore

Keywords: climate, justice, urban greening, history, trees, tree plantings

Author(s): Brandon Miller

Urban geographers have provided critical insights into the dynamics of urban environmentalism, and how these dynamics affect political and scientific understandings of nature. The exclusion of community knowledges and expertise can lead to hegemonic constructs of marginalization, while also creating socioeconomic injustices such as displacement. I argue that these understandings of landscapes and greenspaces have served to remove agency and capacity from the communities that exist within them. Following the onset of climate change, city governments have commonly used trees as a medium to address and mitigate its effects. To this end, I am using urban tree plantings in Franklin Square, Baltimore as a lens into equity. I seek to understand the current frameworks of nature informing tree plantings in the community of Franklin Square. Furthermore, what can the successes and challenges of tree plantings in Franklin Square tell us about efforts to increase tree canopy overall in Baltimore in relation to environmental and climate justice? I have used a combined approach of critical environmental history with the scholarship of urban greening and community-driven stewardship. To gather the history, experiences, and perspectives of Franklin Square, I have been utilizing qualitative geographical methods such as interviews and surveys to capture how the community engages with the greenspaces around them. I have analyzed policy and greening documents through discourse analysis to highlight differences in understandings. With my research, I aim to

help render visible the knowledge and experiences of those in Baltimore to politically inform more equitable and accessible urban greening initiatives.

Monitoring Emergent Tropical Tree Density to Landscape Changes Using High-Resolution Satellite Imagery and Machine Learning: A Case Study with Dipteryx panamensis

Keywords: Tropical Forests, Emergent trees, Remote Sensing

Author(s): Felipe Saad, *UMBC-GES*
Matthew Fagan, *UMBC - GES*
Rohit Mukherjee, *PNNL*

Monitoring tropical tree populations and their response to disturbances is crucial for understanding forest dynamics, the impact of climate change, and species conservation. In this study, we tested the feasibility of employing machine learning techniques on satellite imagery with high resolution (Sentinel 2, 10 meters, and Planetscope, 5 meters) to assess the population density of an emergent tree, Dipteryx panamensis, from 2016 to 2023. This tree, distributed across the lowlands of the San Juan-La Selva Biological Corridor in Costa Rica and Nicaragua, is essential for conserving the critically endangered Great Green Macaw (Ara ambiguus). D. panamensis provides the Great Green Macaw with critical food and nesting resources during the breeding season. Although the forests of northern Costa Rica have been widely cleared and logged, D. panamensis remains relatively abundant inside and outside forests. Here, we mapped and examined the distribution and density patterns of D. panamensis. We employed the Random Forest technique on Sentinel 2 and planetscope data, combined with field data to identify the emergent tree crowns effectively, with accuracy above 90%. In conclusion, machine learning algorithms effectively identify emergent tree individuals in highly diverse tropical forests. This algorithm will be used as input to estimate the mortality of emergent tree populations, which are quite sensitive to fire and fragmentation in some areas. Integrating satellite remote sensing and machine learning techniques are promising approaches for monitoring emergent tropical tree populations and assessing their response to disturbances.

Geography Matters: MAD-AAG 2024

Thank you:

- To our keynote speaker;
- To our presenters;
- To our attendees;
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- To our UMBC hosts (and a shout out to Maggie Holland, Donna Dwyer, Hasnain Ali);
- & to the MAD-AAG Community